

The facet is exactly minimal among all two-level central sections of the regular simplex

Autonomous proof packet for arXiv:2403.13224

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Status. Likely valid substantial partial result. The theorem below proves the conjectured sharp minimum for every central normal having exactly two coordinate values, in every dimension. It does not prove the conjecture for an arbitrary normal.

1 Source question and known boundary

Let

$$\Delta_n = \{x \in \mathbb{R}^{n+1} : x_j \geq 0, \sum_j x_j = 1\}$$

be the regular n -simplex of side length $\sqrt{2}$. For a unit vector $a \perp (1, \dots, 1)$, put

$$T(a) = \Delta_n \cap \{x : \langle a, x \rangle = 0\}.$$

Tang asks whether the central hyperplane parallel to a facet always gives the minimum possible $(n-1)$ -volume [1].

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Question 16. *Can we remove the factor of $1 - o(1)$, so that we can say that $\text{vol}_{n-1}(H_{\text{facet}} \cap \Delta_n)$ is exactly the minimum value of a central section?*

8. ACKNOWLEDGEMENTS

Figure 1: Question 16 on page 17 of the source paper [1].

Tang recalls Webb's probabilistic formula. If $Y_1, \dots, Y_{n+1}$ are independent standard exponential random variables and G_a is the density of $\sum_j a_j Y_j$, then

$$\text{vol}_{n-1} T(a) = \frac{\sqrt{n+1}}{(n-1)!} G_a(0). \tag{1}$$

For the facet normal,

$$a_{\text{fac}} = \left(\sqrt{\frac{n}{n+1}}, -\frac{1}{\sqrt{n(n+1)}}, \dots, -\frac{1}{\sqrt{n(n+1)}} \right),$$

the density is

$$c_n := G_{a_{\text{fac}}}(0) = \left(\frac{n}{n+1} \right)^{n-1/2}. \tag{2}$$

Dirksen proved the sharp bound on the entire chamber with exactly one positive coordinate (and, by changing sign, exactly one negative coordinate), and proved the full conjecture for $n \leq 4$ [2]. The result below is different: it compares all possible sign multiplicities when each sign class is internally constant.

2 Proof intuition

A two-level central normal is forced by the zero-sum and unit-norm constraints once the two multiplicities are fixed. Its positive and negative exponential sums are scaled Gamma variables. Integrating the two Gamma densities at a common positive value gives a closed formula $D_{k,m}$ for $G_a(0)$.

The comparison between adjacent multiplicities then collapses to a one-variable monotonicity fact. Namely,

$$\frac{D_{k+1,m-1}}{D_{k,m}} = \frac{A(k)}{A(m-1)}, \quad A(x) = \left(1 + \frac{1}{x}\right)^{x+1/2}.$$

The function A is strictly decreasing. Moving from the facet multiplicity $k = 1$ toward a balanced split therefore strictly increases the density and, by (1), the section volume. What remains open is controlling normals whose positive or negative coordinates are not all equal.

3 Exact theorem

Call a central normal *two-level* if its coordinates assume exactly two distinct values.

Theorem 1 (Complete two-level classification). *Let $n \geq 2$, put $d = n + 1$, and let $a \in \mathbb{R}^d$ be a two-level unit vector with $\sum_j a_j = 0$. After a permutation and possibly replacing a by $-a$, there are integers $k, m \geq 1$, $k + m = d$, $k \leq m$, such that*

$$a = (\underbrace{\alpha, \dots, \alpha}_k, \underbrace{-\beta, \dots, -\beta}_m), \quad \alpha, \beta > 0.$$

Then

$$G_a(0) = D_{k,m} := \binom{d-2}{k-1} \frac{k^{k-1/2} m^{m-1/2}}{d^{d-3/2}}. \quad (3)$$

For fixed d , the values $D_{k,d-k}$ are strictly increasing for $1 \leq k \leq \lfloor d/2 \rfloor$. Consequently

$$G_a(0) \geq D_{1,n} = \left(\frac{n}{n+1}\right)^{n-1/2} = c_n,$$

with equality if and only if one of the two coordinate values has multiplicity one. Equivalently, among all two-level central sections of Δ_n , the sections parallel to a facet are the unique minimizers, up to the symmetries of the simplex.

Proof. The constraints $k\alpha = m\beta$ and $k\alpha^2 + m\beta^2 = 1$ give

$$\alpha = \sqrt{\frac{m}{kd}}, \quad \beta = \sqrt{\frac{k}{md}}. \quad (4)$$

Let $X \sim \text{Gamma}(k, 1)$ and $Y \sim \text{Gamma}(m, 1)$ be independent. Then $\sum_j a_j Y_j$ has the same law as $\alpha X - \beta Y$. The density of the latter variable at zero is the overlap integral of the densities of αX and βY :

$$\begin{aligned} G_a(0) &= \int_0^\infty \frac{t^{k-1} e^{-t/\alpha}}{\Gamma(k) \alpha^k} \frac{t^{m-1} e^{-t/\beta}}{\Gamma(m) \beta^m} dt \\ &= \frac{\Gamma(d-1)}{\Gamma(k) \Gamma(m)} \frac{\alpha^{m-1} \beta^{k-1}}{(\alpha + \beta)^{d-1}}. \end{aligned}$$

Using (4), one has

$$\alpha = \frac{m}{\sqrt{kmd}}, \quad \beta = \frac{k}{\sqrt{kmd}}, \quad \alpha + \beta = \frac{d}{\sqrt{kmd}}.$$

Substitution, together with $\Gamma(d-1)/(\Gamma(k)\Gamma(m)) = \binom{d-2}{k-1}$, yields exactly (3).

It remains to compare multiplicities. If $1 \leq k < m$ and $k + m = d$, direct cancellation in (3) gives

$$\frac{D_{k+1, m-1}}{D_{k, m}} = \left(1 + \frac{1}{k}\right)^{k+1/2} \left(1 - \frac{1}{m}\right)^{m-1/2} \quad (5)$$

$$= \frac{A(k)}{A(m-1)}, \quad A(x) := \left(1 + \frac{1}{x}\right)^{x+1/2}. \quad (6)$$

We claim that A is strictly decreasing on $(0, \infty)$. Put $t = 1/x$. Then

$$\frac{d}{dx} \log A(x) = \log(1+t) - \frac{t(2+t)}{2(1+t)}.$$

If

$$H(t) = \frac{t(2+t)}{2(1+t)} - \log(1+t),$$

then $H(0) = 0$ and

$$H'(t) = \frac{t^2}{2(1+t)^2} > 0 \quad (t > 0).$$

Thus $(\log A)' < 0$. For every $1 \leq k < \lfloor d/2 \rfloor$, the adjacent pair $k, k+1$ stays inside the half-range and one has $k < m-1$, hence $A(k) > A(m-1)$ and (5) is strictly greater than one. Therefore $D_{k, d-k}$ is strictly increasing on that range. Its minimum is $k=1$, where (3) becomes

$$D_{1, n} = \frac{n^{n-1/2}}{(n+1)^{n-1/2}} = c_n.$$

The equality and geometric statements follow from (1) and from permutation/sign symmetry. $\square$

4 Verification and limitations

The accompanying script independently checks the Gamma overlap integral, formula (3), and ratio (5) at high precision for a representative collection of multiplicities, and checks monotonicity through $d=200$. These calculations are sanity checks, not ingredients of the proof.

The theorem does not imply that the minimum inside every fixed sign chamber is two-level. In fact, Dirksen explicitly warns that this tempting reduction is false in higher dimensions [2, Remark

3.4 and Lemma 3.6]. His balancing proposition makes all coordinates on one sign equal but leaves the other sign ratios free. Controlling those remaining ratios is the open core of Tang’s Question 16.

Several stronger routes were tested. Numerical quadrature and global optimization in dimensions 5 through 11 returned the facet normal to numerical precision, and a pointwise comparison on Tang’s positive contour also survived randomized tests. A sharpened Cauchy–Schwarz step using $a \perp (1, \dots, 1)$ yields a genuine positive deficit, but converting that deficit into the exact facet constant requires a new rational moment inequality. None of these numerical or unfinished observations is claimed as a proof.

5 Bounded novelty check and review recommendation

The 11 August 2026 search covered the run’s registry/solution/attempt indexes, exact arXiv-id and keyword queries, Dirksen’s arXiv:1509.06408, and the later arXiv:2505.00944 and arXiv:2606.07163. Dirksen’s paper contains the closest known analysis, including the one-sign theorem and examples involving two-level normals, but the all-multiplicity monotonicity in Theorem 1 was not located. The two later papers concern maximum-section or reverse-Hölder questions rather than this exact minimum. Novelty confidence is moderate: the result may be folklore because the Gamma calculation is short.

Human review should focus on the Gamma scaling in (3) and the strictness range in (5). The packet should remain classified as a partial result unless the arbitrary multi-level case is separately proved.

References

- [1] C. Tang, *Simplex slicing: an asymptotically-sharp lower bound*, arXiv:2403.13224, 2024.
- [2] H. Dirksen, *Sections of the regular simplex – volume formulas and estimates*, arXiv:1509.06408; Math. Nachr. 290 (2017), 2567–2584.